



1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE SUPPLIER

1.1 Product identifiers

Product Name	: Chlorfenapyr 19% + Lufenuron 5% W/V SC
Brand	: Sunjoy Agro
Product Code	: CIPAC No. Chlorfenapyr: 570; Lufenuron: 704
CAS Number	: Chlorfenapyr: 122453-73-0; Lufenuron: 103055-07-8
Chemical name of a.i.	: Chlorfenapyr: 4-bromo-2-(4-chlorophenyl)-1-ethoxymethyl-5-trifluoromethyl-1H-pyrrole-3-carbonitrile (IUPAC) 4-bromo-2-(4-chlorophenyl)-1-(ethoxymethyl)-5-(trifluoromethyl)-1H-pyrrole-3-carbonitrile (CA) Lufenuron: (RS)-1-[2,5-dichloro-4-(1,1,2,3,3,3-hexafluoropropoxy)phenyl]-3-(2,6-difluorobenzoyl)urea (IUPAC) N-[[[2,5-dichloro-4-(1,1,2,3,3,3-hexafluoropropoxy)phenyl]amino]carbonyl]-2,6-difluorobenzamide (CA)
Chemical formulation	: Chlorfenapyr: C ₁₅ H ₁₁ BrClF ₃ N ₂ O; Lufenuron: C ₁₇ H ₈ Cl ₂ F ₈ N ₂ O ₃
Formulation	: Suspension concentrate (SC)

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses	: Insecticide
Uses advised against	: See chapter 16 for a general overview.

1.3 Details of the supplier of the safety data sheet

Company	: Ningbo Sunjoy Agrosience Co., Ltd.
Address	: No. 1165 Beihai Road, Chemical Industry Zone of Ningbo, Zhenhai District, Ningbo, Zhejiang Province, China.
Telephone	: +86-574-87774749 87760080
Fax	: +86-574-87774751

NINGBO SUNJOY AGROSCIENCE CO., LTD.
宁波三江益农化学有限公司

1.4 Emergency telephone number

Emergency Phone #	: +86-574-87774749
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2. HAZARD IDENTIFICATION

2.1 Classification of the substance or mixture:

Classification according to Regulation (EC) No 1272/2008:

Acute toxicity, inhalation - Category 4

Hazardous to the aquatic environment, acute hazard - Category 1

Hazardous to the aquatic environment, long-term hazard - Category 1

2.2 GHS label elements:

Hazard Pictograms:



Signal word:

Warning

Hazard statement(s):

H332 : Harmful if inhaled
H400 : Very toxic to aquatic life
H410 : Very toxic to aquatic life with long lasting effects

Precautionary statement(s):**Prevention**

P261 : Avoid breathing vapours.
P271 : Use only outdoors or in a well-ventilated area.
P273 : Avoid release to the environment.

Response

P304 + P340 : IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P312 : Call a poison center/doctor if you feel unwell.
P391 : Collect spillage.
Disposal
P501 : Dispose of container in accordance with local regulations

2.3 Hazards not otherwise classified or not covered by GHS:

No specific dangers known, if the regulations/notes for storage and handling are considered.

3. COMPOSITION/INFORMATION ON INGREDIENTS**3.1 Active ingredient(s):**

Common name : Chlorfenapyr; Lufenuron
Formula : Chlorfenapyr: $C_{15}H_{11}BrClF_3N_2O$;
Lufenuron: $C_{17}H_8Cl_2F_8N_2O_3$
Molecular weight : Chlorfenapyr: 407.6; Lufenuron: 511.2
CAS No. : Chlorfenapyr: 122453-73-0; Lufenuron: 103055-07-8
Content : Chlorfenapyr 19% W/V; Lufenuron 5% W/V

3.2 Other ingredients:

Ingredients not precisely identified are proprietary or non-hazardous. Values are not product specifications.

4. FIRST AID MEASURES**4.1 Description of first aid measures:****4.1.1 Following Inhalation:**

Move person to fresh air. If person is not breathing, call an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice.

4.1.2 Following Skin Contact:

Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice.

4.1.3 Following Eye Contact:

Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice.

4.1.4 Following Ingestion:

Provided the patient is conscious, give 1 or 2 glasses of water to drink. Vomiting should only be induced under the direction of a physician or a poison control center. If spontaneous vomiting

occurs, have victim lean forward with head down to avoid breathing in of vomitus, rinse mouth and administer water. Immediately transport victim to an emergency facility.

4.2 Most important symptoms and effects, both acute and delayed

No special symptoms based on tests on animals.

4.3 Indication of any immediate medical attention and special treatment needed

There is no specific antidote if this product is ingested. Treat symptomatically.

5. FIRE FIGHTING MEASURES

5.1 Extinguish media:

Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

5.2 Special hazards arising from the substance or mixture

During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

5.3 Advice for fire-fighters

Wear full protective clothing and self-contained breathing apparatus. Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated buildings, area, and equipment until decontaminated. Water runoff can cause environmental damage. If water is used to fight fire, dike and collect runoff.

5.4 Further information

No data available

6. ACCIDENTAL RELEASE MEASURES

6.1 Personal Precautions, protective equipment and emergence procedures

Control the spill at its source. Contain the spill to prevent from spreading or contaminating soil or from entering sewage and drainage systems or any body of water. Clean up spills immediately, observing precautions in Protective Equipment Section. Sweep up material and place in a compatible disposal container. Scrub area with hard water detergent (e.g. commercial products such as Tide, Joy, Spic and Span). Pick up wash liquid with additional absorbent and place into compatible disposal container. Once all material is cleaned up and placed in a disposal container, seal container and arrange for disposition.

6.2 Environmental Precautions

Toxic to aquatic life with long lasting effects.

6.3 Methods and Material for containment and cleaning up

Dispose of product containers, waste containers, and residues according to local, state, and federal health and environmental regulations.

6.4 Additional advices

Nothing

7. HANDLING AND STORAGE

7.1 Precautions for safe handling

Keep out of reach of children. When opening the container and preparing use, wear cotton overalls buttoned to the neck and wrist and a washable hat and elbow-length PVC gloves. Wash hands after use. After each day's use, wash gloves and contaminated clothing.

7.2 Conditions for safe storage, including any incompatibilities

7.2.1 Technical measures and storage conditions

Store the material in a well-ventilated, secure area out of reach of children and domestic animals. Do not store near sources of heat including steam lines. Do not store food, beverages or tobacco products in the storage area. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

7.2.2 Packaging materials

Not defined

7.2.3 Requirements for storage rooms and vessels

General requirements of storage

7.3 Specific end use(s)

None

8. EXPOSURE CONTROL/PERSONAL PROTECTION

8.1 Control parameters

Components with workplace control parameters

Contains no substances with occupational exposure limit values.

8.2 Appropriate engineering controls

Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of workday.

8.3 Individual protection measures

Personal protective equipment

Ingestion: Prevent eating, drinking, tobacco usage and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

Eye Contact: Where eye contact is likely, use chemical splash goggles.

Skin Contact: Where contact is likely, wear chemical-resistant (such as nitrile or butyl) gloves, coveralls, socks and chemical-resistant footwear. For overhead exposure, wear chemical-resistant headgear.

Inhalation: Use process enclosures, local exhaust ventilation, or other engineering controls to keep airborne levels below exposure limits. A NIOSH-certified combination air-purifying respirator with an N, P or R 95 or HE class filter and an organic vapor cartridge may be permissible under certain circumstances where airborne concentrations are expected to exceed exposure limits. Protection provided by air-purifying respirators is limited. Use a pressure demand atmosphere-supplying respirator if there is any potential for uncontrolled release, exposure levels are not known, or under any other circumstances where air-purifying respirators may not provide adequate protection.

9. PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Physical state	: Liquid
Color	: Off-white
Odor	: Negligible
M.p.	: No data available
Flammability	: Non-flammable
Lower and upper explosion limit/flammability limit	: No data available
Flash point	: > 100°C
Auto-ignition temperature	: No data available

Decomposition temperature	: No data available
pH	: 5.0 – 8.0
Kinematic viscosity	: No data available
Solubility in water	: Suspension
Solubility in organic solvents	: No data available
Partition coefficient: n-octanol/water (log value)	: No data available
Vapour pressure	: No data available
Density/relative density	: 1.12 ± 0.02 g/cm ³ (20°C)
Relative vapour density	: No data available
Explosive properties	: Not explosive
Oxidizing properties	: Not oxidizing

9.2 Other safety information

Further safety related physical-chemical data are not known.

10. STABILITY AND REACTIVITY

10.1 Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

10.2 Chemical Stability

Stable under normal use and storage conditions.

10.3 Possibility of hazardous reaction

Can decompose at high temperatures forming toxic gases.

10.4 Conditions to avoid

Avoid humidity, temperature extremes.

10.5 Incompatible materials

Strong acids, alkalies, and oxidizing agents.

10.6 Hazardous decomposition products

Other decomposition products: No data available
In the event of fire: see section 5

11. TOXICOLOGICAL INFORMATION

11.1 Information on the likely routes of exposure

Acute Oral Toxicity	Rat: 2000 mg/kg <LD ₅₀ < 5000 mg/kg.
Acute Dermal Toxicity	Rat: LD ₅₀ > 2000 mg/kg.
Acute Inhalation Toxicity	Rat: LC ₅₀ (4 h) > 2.029 mg/L.
Eye Irritation	Rabbit: Not eye irritant.
Skin Irritation	In vitro in RHE method: Not skin irritant.
Sensitization	LLNA method: Not a skin sensitizer.

11.2 Symptoms related to the physical, chemical and toxicological characteristics

No specific symptoms were found based on animals tests.

11.3 Delayed and immediate effects and also chronic effects from short- and long-term exposure

Germ cell mutagenicity
Not mutagenic

Carcinogenicity

Not carcinogenic

Reproductive toxicity

No developmental effects and no reproductive effects

Specific target organ toxicity

None

To the best of our knowledge, the chemical, physical, and toxicological properties have not been thoroughly investigated.

11.4 Numerical measure of toxicity

NOEL No data available

ADI No data available

12. ECOLOGICAL INFORMATION**12.1 Ecotoxicity****For Chlorfenapyr Technical**

Birds: Acute oral LD₅₀ for mallard ducks 10, bobwhite quail 34 mg/kg. Dietary LC₅₀ (8 d) for mallard ducks 9.4, bobwhite quail 132 mg/kg diet.

Fish: LC₅₀ (48 h) for carp 0.5 mg/l; (96 h) for rainbow trout 0.00744, bluegill sunfish 0.0116 mg/l.

Daphnia: LC₅₀ (96 h) 0.00611 mg/l.

Algae: EC₅₀ for *Selenastrum capricornutum* 0.132 mg/l.

Bees: LD₅₀ 0.2 µg/bee.

For Lufenuron Technical:

Birds: Acute oral LD₅₀ for bobwhite quail and mallard ducks >2000 mg/kg. Dietary LC₅₀ (8 d) for bobwhite quail and mallard ducks >5200 mg/kg diet.

Fish: LC₅₀ (96 h) for rainbow trout >73, carp >63, bluegill sunfish >29, catfish >45 mg/l.

Daphnia: EC₅₀ (48 h) 0.0013 mg/l.

Algae: EC₅₀ (72 h) for *Selenastrum capricornutum* 8.8 mg/l.

Bees: LD₅₀ (oral) >197 µg/bee; LD₅₀ (contact) >200 µg/bee.

Worms: LC₅₀ (14 d) for earthworms >1000 mg/kg.

12.2 Persistence and degradability**For Chlorfenapyr Technical**

In soil, chlorfenapyr is the major residue. Debromination to a less toxic metabolite is the primary route of degradation; dealkylation is not a primary route of degradation in soil. In water, DT₅₀ (direct photodegradation) 4.8–7.5 d; stable to hydrolysis at pH 4, 7 and 9.

For Lufenuron Technical

Lufenuron was rapidly degraded in biologically active soils under aerobic conditions. DT₅₀ 9.4–83.1 d.

12.3 Bioaccumulative potential for technical

No data available.

12.4 Mobility in soil**For Chlorfenapyr Technical**

K_{oc} >10 000 ml/g, indicating chlorfenapyr is likely to be strongly bound in soils.

For Lufenuron Technical

Lufenuron showed a very strong adsorption onto soil particles: K_{oc} (mean value) 38 mg/g o.c.

12.5 Other adverse effects

An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Toxic to aquatic life.

13. DISPOSAL CONSIDERATIONS

13.1 Container disposal method

Do not reuse this container for any purpose. This is a recyclable container, and is to be disposed of at a container collection site. Contact your local distributor/dealer or municipality for the location of the nearest collection site. Before taking the container to the collection site:

1. Triple- or pressure-rinse the empty container. Add the rinsings to the spray mixture in the tank.
2. Make the empty, rinsed container unsuitable for further use.

If there is no container collection site in your area, dispose of the container in accordance with provincial requirements.

13.2 Pesticide disposal method

For information on disposal of unused, unwanted product, contact the manufacturer or the provincial regulatory agency. Contact the manufacturer and the provincial regulatory agency in case of a spill, and for clean-up of spills.

14. TRANSPORTATION INFORMATION

Ground Transport (ADR/RID/AND/DOT(US))

- | | |
|-------------------------------|--|
| 14.1 Proper shipping name | : Environmentally hazardous substance, liquid, n.o.s.
(Chlorfenapyr, Lufenuron) |
| 14.2 UN Number | : 3082 |
| 14.3 Hazard class | : 9 |
| 14.4 Packing Group | : III |
| 14.5 Environm. Hazardous mark | : YES |

Water Transport (IMDG)

- | | |
|---------------------------|--|
| 14.1 Proper shipping name | : Environmentally hazardous substance, liquid, n.o.s.
(Chlorfenapyr, Lufenuron) |
| 14.2 UN Number | : 3082 |
| 14.3 Hazard class | : 9 |
| 14.4 Packing Group | : III |
| 14.5 Marine pollutant | : YES |

Air Transport (IATA)

- | | |
|-------------------------------|--|
| 14.1 Proper shipping name | : Environmentally hazardous substance, liquid, n.o.s.
(Chlorfenapyr, Lufenuron) |
| 14.2 UN Number | : 3082 |
| 14.3 Hazard class | : 9 |
| 14.4 Packing Group | : III |
| 14.5 Environm. Hazardous mark | : YES |

15. REGULATORY INFORMATION

15.1 Classification

EPCRA SARA Title III Classification:

- | | | |
|--|---|------|
| Section 302 Notification and information | : | None |
| Section 313 Toxic chemical | : | None |

California Proposition 65:
Not Applicable

RCRA Hazardous Waste Classification (40 CFR 261):

Not Applicable

15.2 Risk Phrases

- R20 Harmful by inhalation
- R50 Very toxic to aquatic organisms
- R53 May cause long-term adverse effects in the aquatic environment

15.3 Safety Phrases

- S23 Do not breathe gas/fumes/vapour/spray
- S35 This material and its container must be disposed of in a safe way
- S51 Use only in well-ventilated areas
- S61 Avoid release to the environment. Refer to special instructions/safety data sheets
- S63 In case of accident by inhalation: remove casualty to fresh air and keep at rest

16. OTHER INFORMATION

SDS Number : SJ570+704/24%SC
Prepared by : Pony Fang

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. The data do not describe the product's properties (product specification). Neither should any agreed property nor the suitability of the product for any specific purpose be deduced from the data contained in the safety data sheet. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.